



CANDIDATE
NAME

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CENTRE
NUMBER

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CANDIDATE
NUMBER

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9702/33

October/November 2021

2 hours

You will need: The materials and apparatus listed in the confidential instructions

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You will be allowed to work with the apparatus for a maximum of 1 hour for each question.
- You should record all your observations in the spaces provided in the question paper as soon as these observations are made.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 40.
- The number of marks for each question or part question is shown in brackets [].

For Examiner's Use	
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You may not need to use all of the materials provided.

1 In this experiment, you will investigate combinations of resistors in an electrical circuit.

(a) Fig. 1.1. shows an electrical circuit.

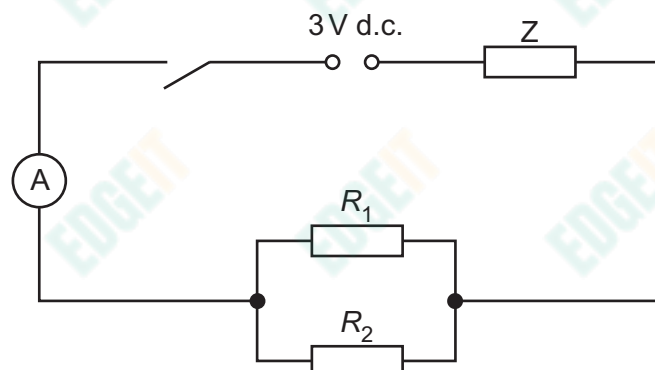


Fig. 1.1

- Set up the circuit shown in Fig. 1.1 using $R_1 = 33\ \Omega$ and $R_2 = 82\ \Omega$.
- Calculate $\frac{R_1 R_2}{(R_1 + R_2)}$.

$$\frac{R_1 R_2}{(R_1 + R_2)} = \dots\dots\dots \Omega$$

- Close the switch.
- Record the ammeter reading I .

$$I = \dots\dots\dots$$

- Open the switch.

[1]

- (b) Use six different pairs of resistors to provide six different values of $\frac{R_1 R_2}{(R_1 + R_2)}$.

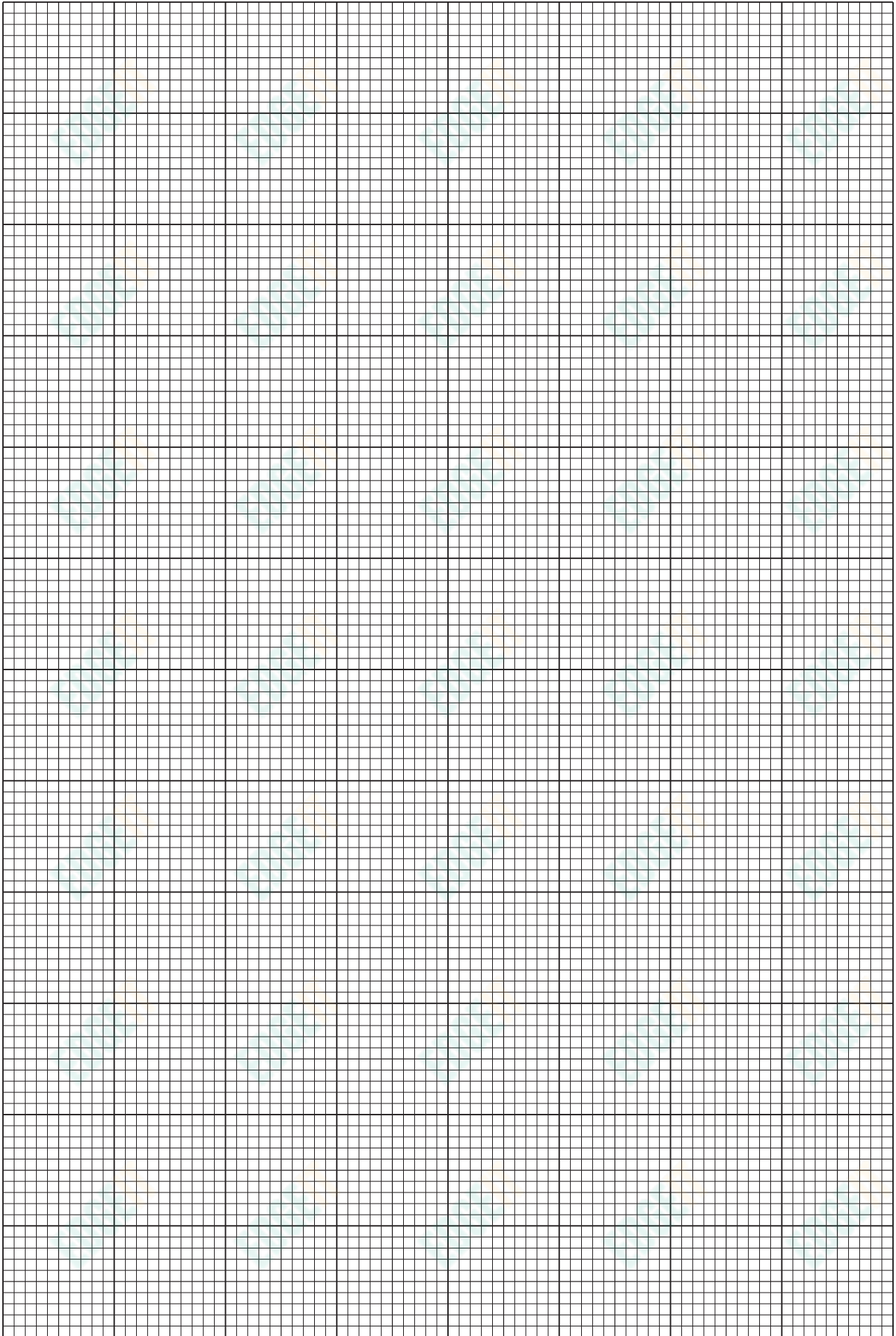
For each arrangement, record R_1 , R_2 and I in a table. Include values of $\frac{R_1 R_2}{(R_1 + R_2)}$ and $\frac{1}{I}$ in your table.

- (c) (i) Plot a graph of $\frac{1}{I}$ on the y-axis against $\frac{R_1 R_2}{(R_1 + R_2)}$ on the x-axis. [10]
 (ii) Draw the straight line of best fit. [3]
 (iii) Determine the gradient and y-intercept of this line. [1]

gradient =

y-intercept =

[2]



- (d) (i) It is suggested that the quantities I and $\frac{R_1 R_2}{(R_1 + R_2)}$ are related by the equation

$$\frac{1}{I} = P \left[\frac{R_1 R_2}{(R_1 + R_2)} \right] + Q$$

where P and Q are constants.

Using your answers to (c)(iii), determine the values of P and Q .
Give appropriate units.

$P =$

$Q =$

[2]

- (ii) The constants P and Q are related to the electromotive force (e.m.f.) E of the power supply and the resistance Z of resistor Z by

$$P = \frac{1}{E} \text{ and } Q = \frac{Z}{E}.$$

Determine the values of E and Z .
Give appropriate units.

$E =$

$Z =$

[1]

[Total: 20]

You may not need to use all of the materials provided.

2 In this experiment, you will investigate the time taken for filter papers to fall in air.

(a) (i) • You have been provided with filter papers of two different sizes.

Take one sheet of the **smaller** filter paper.

• The diameter of one sheet of filter paper is d , as shown in Fig. 2.1.

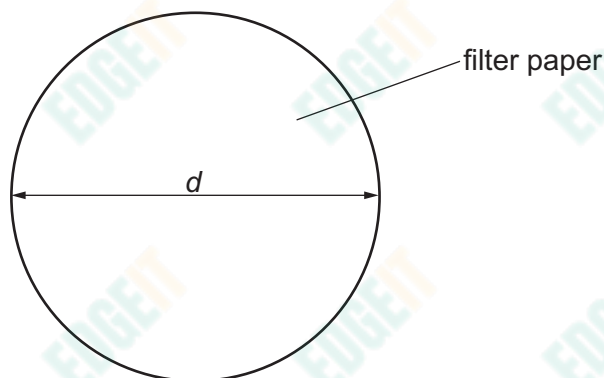


Fig. 2.1

Measure and record d .

$d = \dots\dots\dots$ cm [2]

(ii) Calculate the area A of the filter paper using

$$A = \frac{\pi d^2}{4}.$$

$A = \dots\dots\dots$ cm² [1]

(iii) Justify the number of significant figures that you have given for your value of A .

.....

 [1]

- (b) (i) • Set up the apparatus as shown in Fig. 2.2.

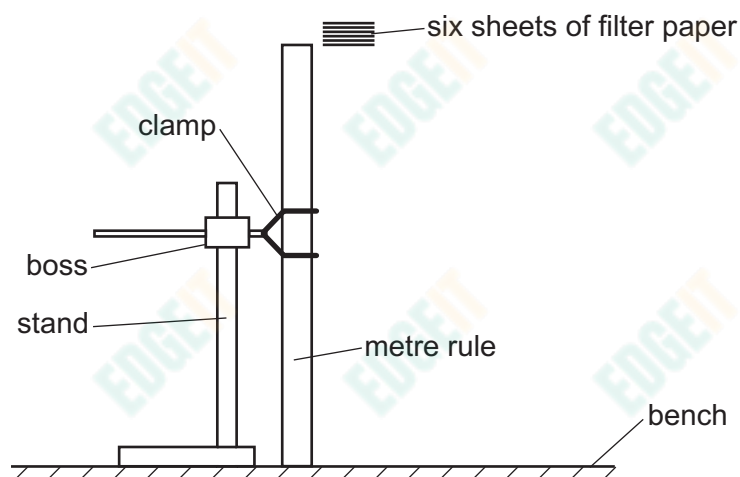


Fig. 2.2

- Hold the six sheets of the smaller filter paper at the top of the metre rule, as shown in Fig. 2.2.
- Release the filter papers and start the stop-watch.
- The time between release and the filter papers hitting the bench is t .
Measure and record t .

$t = \dots\dots\dots$ s [2]

- (ii) Estimate the percentage uncertainty in t . Show your working.

percentage uncertainty = $\dots\dots\dots$ [1]

- (iii) Measure and record the total mass m of the sheets of smaller filter paper.

$m = \dots\dots\dots$ [1]

(c) (i) Repeat (a)(i) and (a)(ii) using one of the **larger** sheets of filter paper.

$$d = \dots\dots\dots \text{ cm}$$

$$A = \dots\dots\dots \text{ cm}^2$$

[1]

(ii) Using two sheets of the larger filter paper, repeat (b)(i) and (b)(iii).

$$t = \dots\dots\dots \text{ s}$$

$$m = \dots\dots\dots$$

[1]

- (d) It is suggested that the relationship between t , m and A is

$$kt = mA$$

where k is a constant.

- (i) Using your data, calculate two values of k .

first value of k =

second value of k =

[1]

- (ii) Explain whether your results support the suggested relationship.

.....

 [1]

(e) (i) Describe four sources of uncertainty or limitations of the procedure for this experiment.

1.
.....
2.
.....
3.
.....
4.
.....

[4]

(ii) Describe four improvements that could be made to this experiment. You may suggest the use of other apparatus or different procedures.

1.
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2.
.....
3.
.....
4.
.....

[4]

[Total: 20]

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